

Advanced Topics to Wireless Communications

- The report accompanying the assignment must be submitted in PDF format. In addition, the simulation code, written in MATLAB/Python, must be submitted along with the report.
- The assignment must be submitted to eLearning by June 30, 2026..
- Contact email for questions and clarifications: padiaman@auth.gr and tegosoti@auth.gr.

Unmanned aerial vehicles (UAVs) are emerging as flexible aerial communication platforms that can provide on-demand coverage, support temporary hotspots, and enhance wireless connectivity in areas where terrestrial infrastructure is limited or overloaded. A key advantage of UAV-based communication systems is the additional design freedom offered by their controllable altitude and antenna configuration, which strongly affect both coverage and link quality. In this assignment, you will study how the joint selection of UAV altitude and antenna beamwidth influences system performance under different multiuser communication models. Based on the paper “Joint Altitude and Beamwidth Optimization for UAV-Enabled Multiuser Communications” [1], you will analyze the underlying tradeoffs, derive the main mathematical expressions, interpret the reported results, and explore how these insights can be extended to broader UAV wireless network scenarios.

1. **(1 point)** Describe the fly-hover-and-communicate protocol and explain why the ground area is partitioned into hexagonal cells.
2. **(1 point)** Why is the fly-hover-and-communicate protocol practical, even if it may be suboptimal? How would fading, interference, or limited UAV energy change the problem? How would the conclusions differ in dense urban UAV networks instead of rural line-of-sight (LoS) environments? Would the same altitude/beamwidth trends still hold if multiple UAVs cooperated?
3. **(2 points)** Based on the UAV-enabled communication model of the paper, re-derive one of the following expressions from first principles: (i) multicast throughput, (ii) broadcast sum rate, or (iii) uplink multiple access rate. Start from the channel model and antenna model, and clearly justify: (a) the coverage geometry, (b) the integration limits, (c) the user distribution assumptions.
4. **(2 points)** Reproduce the numerical results (figures) presented in the paper.
5. **(2 points)** Select one communication model (multicast, broadcast, or multiple access) and consider a geographical area of size A with user density ρ . For different target rates per user R_{target} , determine the minimum number of UAVs required to guarantee that all users achieve at least the corresponding rates. Also, reproduce the results when the user density increases by 20%. Present the results in a figure and comment this.
6. **(2 points)** Answer the following conceptual questions: (a) Why does increasing the UAV altitude improve performance in the multicast case but degrade it in the broadcast case? (b) Why is the uplink multiple access performance independent of the UAV altitude? (c) Explain the fundamental trade-off between antenna beamwidth and communication rate.

[1] H. He, S. Zhang, Y. Zeng and R. Zhang, “Joint Altitude and Beamwidth Optimization for UAV-Enabled Multiuser Communications,” in *IEEE Communications Letters*, vol. 22, no. 2, pp. 344-347, Feb. 2018, doi: 10.1109/LCOMM.2017.2772254.